LLC isolation stage

File: llc.kicad_sch

Primary aux supply, CA-IS3742 isolator,

File: pri_aux.kicad_sch

Isolated aux flyback

File: iso_pwr.kicad_sch

4SBB regulator

File: sbb.kicad_sch

Secondary aux supplies

File: sec_aux.kicad_sch

STM32G474RBT6 MCU

File: mcu.kicad_sch

Sensing

File: sense.kicad_sch

Isolated CAN

File: can.kicad_sch

Address switch, programming pads, ARM SW

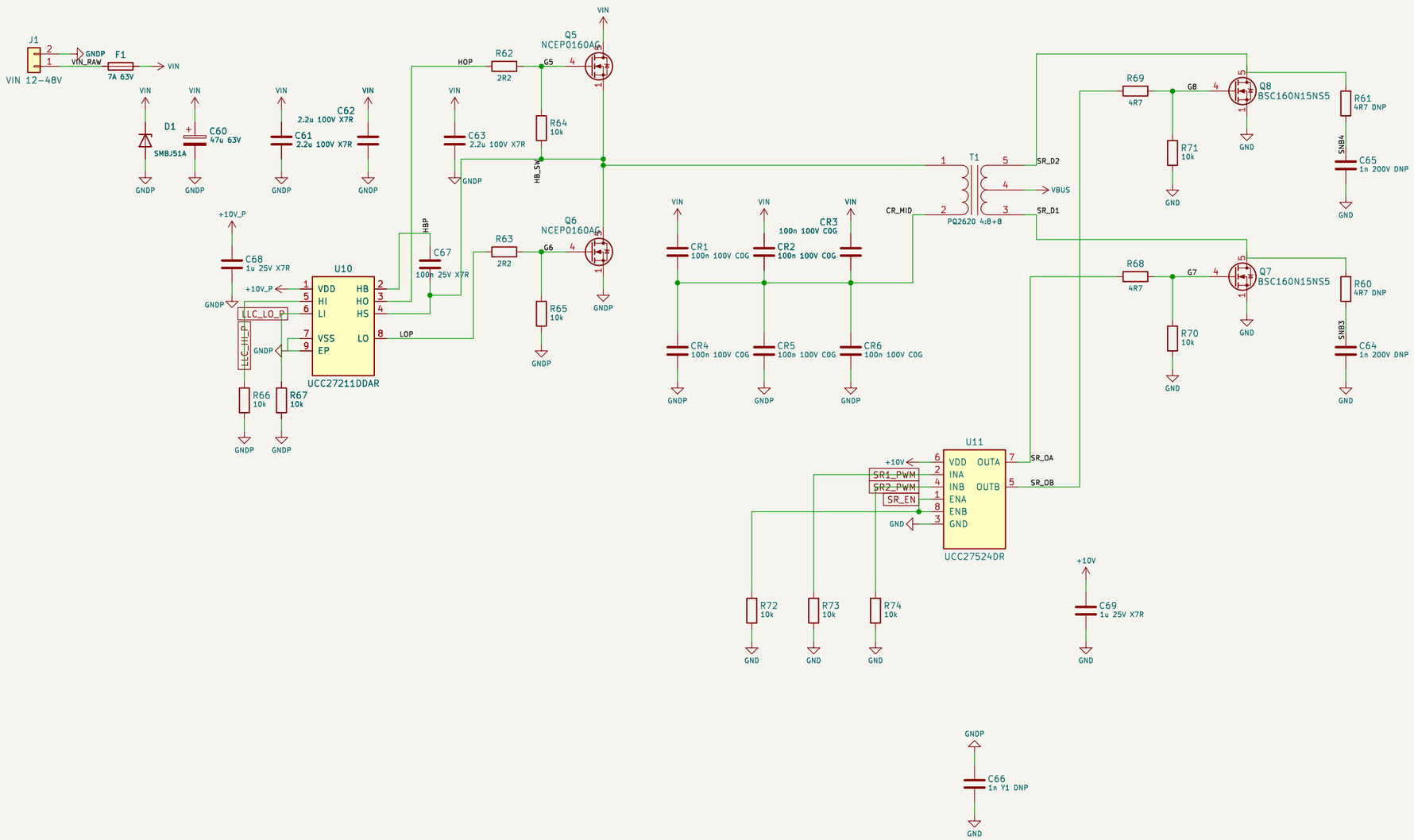
File: io.kicad_sch

Sheets connect through global labels and power symbols
Isolated 12-48 V -> 1.5-48 V / 5 A / 50 W; STM32G474RBT6; isolated CAN

Sheet: /
File: DCDC_LLC_4SBB_G474.kicad_sch

Title: DCDC LLC + 4SBB G474 — top level

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10

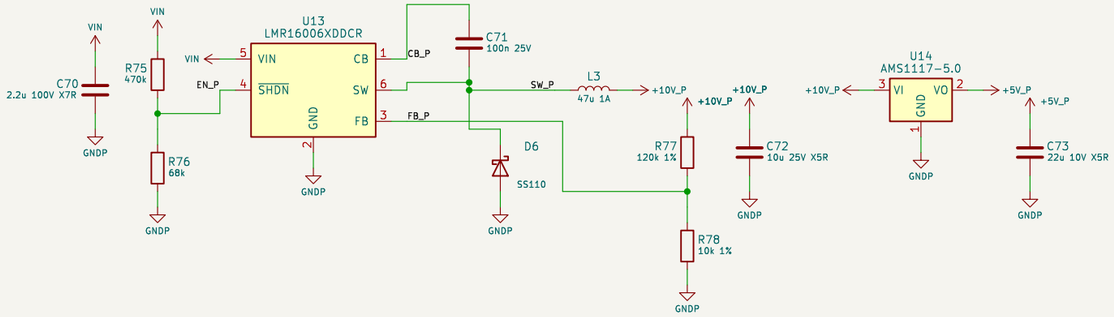


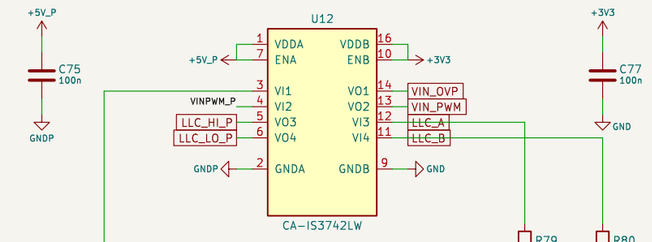
Sheet 2: LLC isolation stage – half-bridge, PQ2620, synchronous rectifier
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

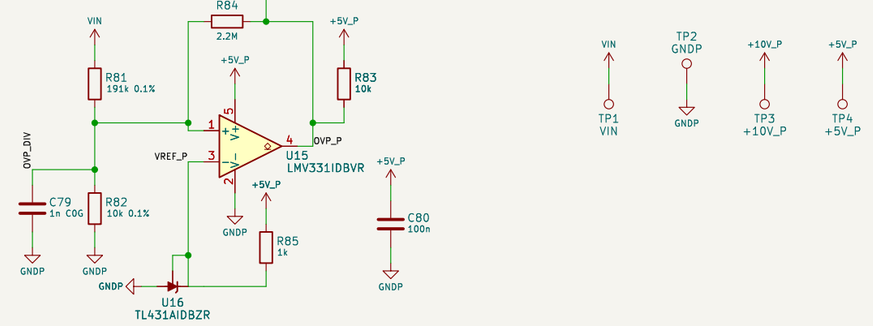
Sheet: /LLC isolation stage/
File: llc.kicad_sch

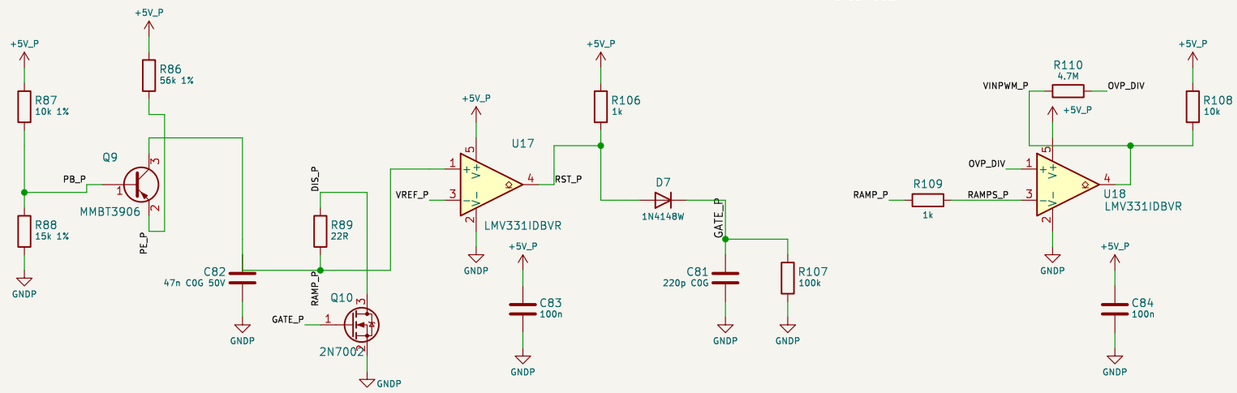
Title: LLC isolation stage – half-bridge, PQ2620, synchronous rectifier

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/10









INPUT OVP (primary side)
Trip = 2.495 V x (1 + 191k/10k) = 50.2 V rising
Hysteresis ~0.4 V (R84 2.2M)
OVP_P high = over-voltage -> U12 VI1 -> VO1
-> VIN_OVP -> PA15 (HRTIM1_FLT2 / EXTI15)

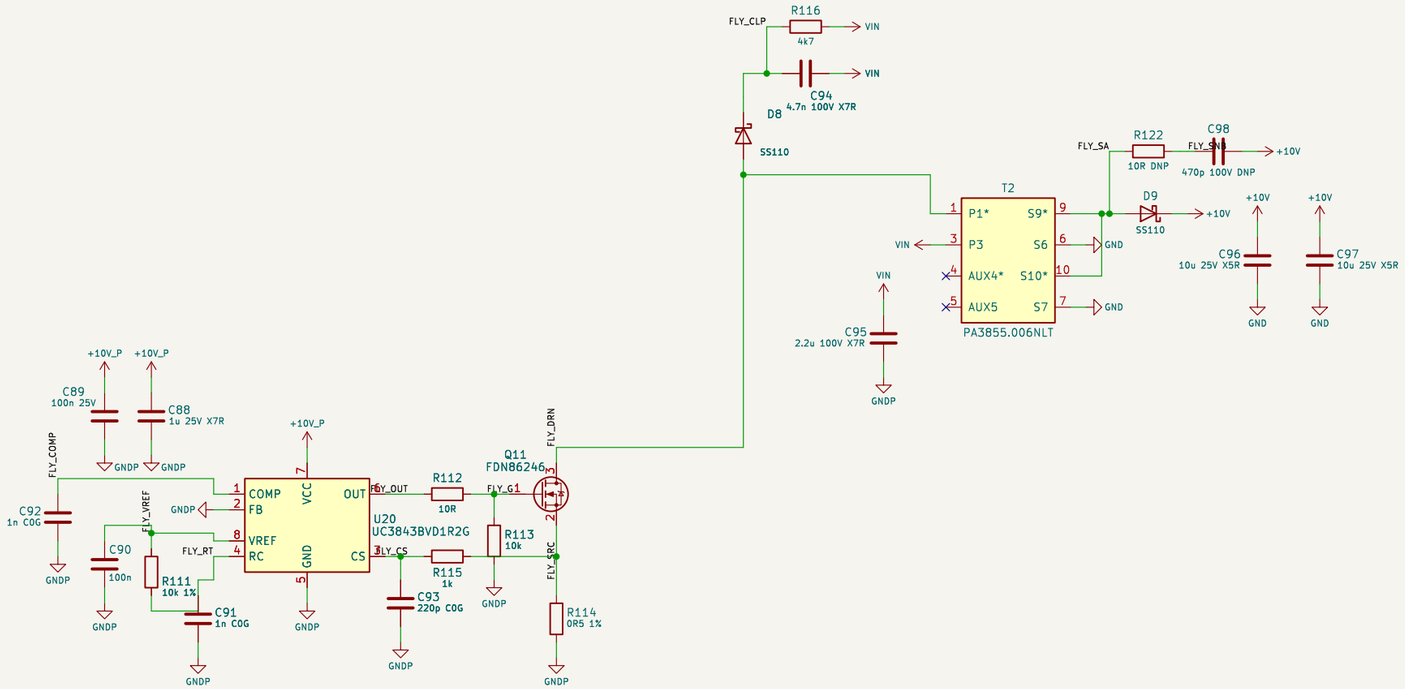
VIN MONITOR (primary side, monitoring only)
Ramp 0 -> 2.495 V in ~5 ms (Q9 ~24 uA into C82 47n C0G), reset ~20 us by U17/D7/C81/Q10
U18: VINPWM_P high while RAMP < OVP_DIV (R110 hysteresis on the divider side) -> U12 VI2 -> VO2 -> VIN_PWM -> PA2 (TIM15_CH1 PWM input)
VIN = 50.47 V x (t_high − t_reset) / (period − t_reset) − 0.20 V, t_reset ~20 us; calibrate once

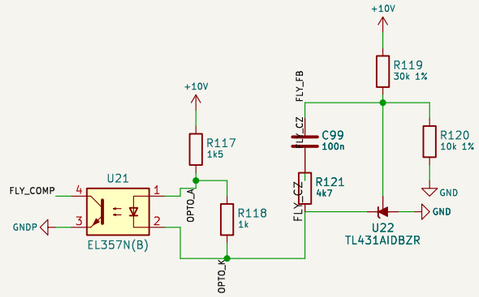
Sheet 3: Primary aux supply, CA-IS3742 isolator, input OVP and VIN monitor
DCDC LLC + 4SBB, 12-48 V in, 1.5-48 V / 5 A out, STM32G474

Sheet: /Primary aux supply, CA-IS3742 isolator, /
File: pri_aux.kicad_sch

Title: Primary aux supply, CA-IS3742 isolator, input OVP and VIN monitor

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/10





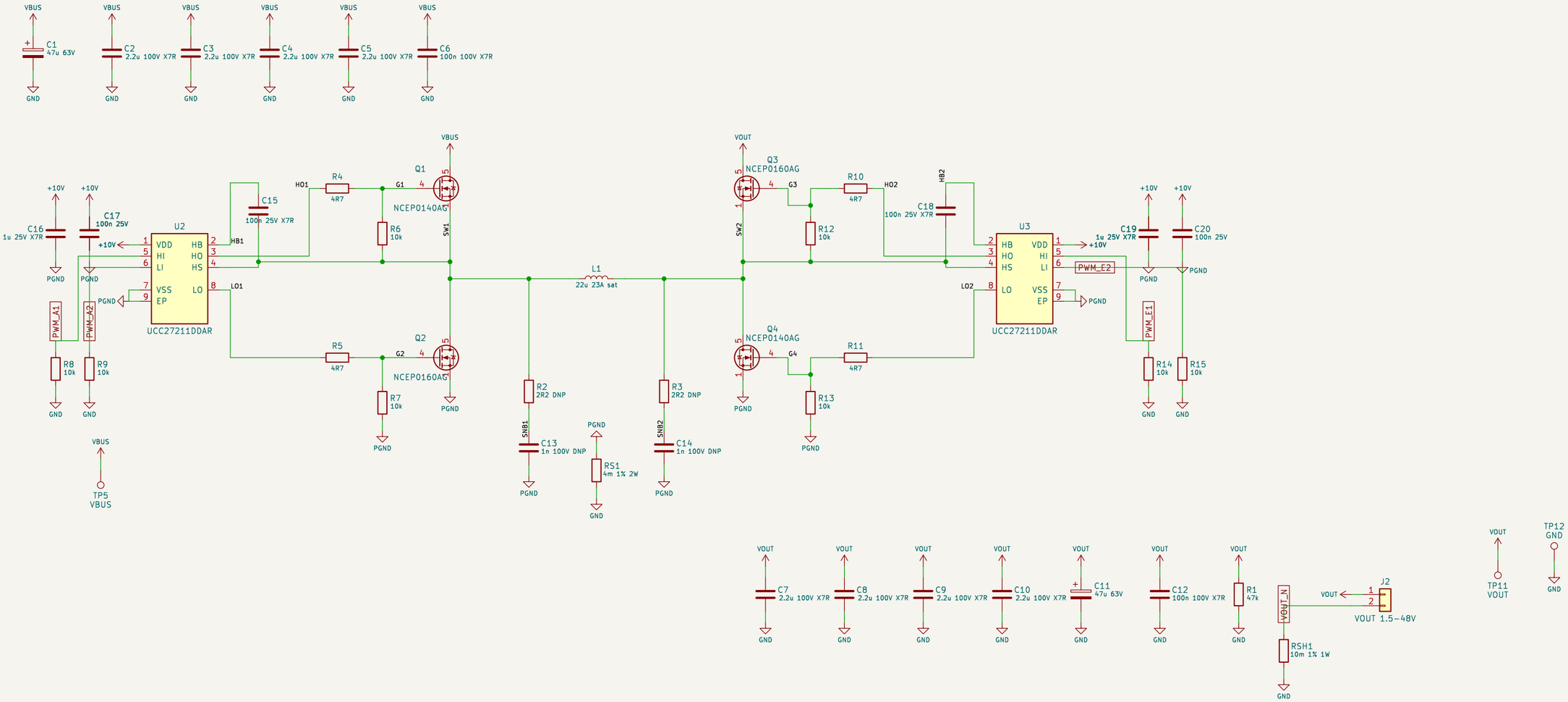
Isolated aux flyback: VIN 12–48 V –> +10 V secondary (1.35 W typ, 3 W design, ~170 kHz DCM).
UC3843B current mode, VCC = +10V_P (U13). Ipk limit 1.0 V / R114 = 2.0 A. RCD clamp D8/R116/C94: Vds < 85 V at 50 V in (150 V FET).
PA3855.006NLT: Np:Ns 1.33:1, Lp 21 uH; secondaries 9–6 and 10–7 in parallel; aux 4–5 not used. Hi-pot 2250 Vdc.
Regulation on the secondary: TL431 U22 (+10V = 2.495 x (1 + R119/R120) = 9.98 V) drives the EL357N LED; the opto
transistor pulls COMP down (FB grounded). Type-2 compensation R121/C99 (zero 340 Hz). Check the step response on the bench.

Sheet 4: Isolated aux flyback – UC3843B, PA3855, TL431 + optocoupler
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

Sheet: /Isolated aux flyback/
File: iso_pwr.kicad_sch

Title: Isolated aux flyback – UC3843B, PA3855, TL431 + optocoupler

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/10

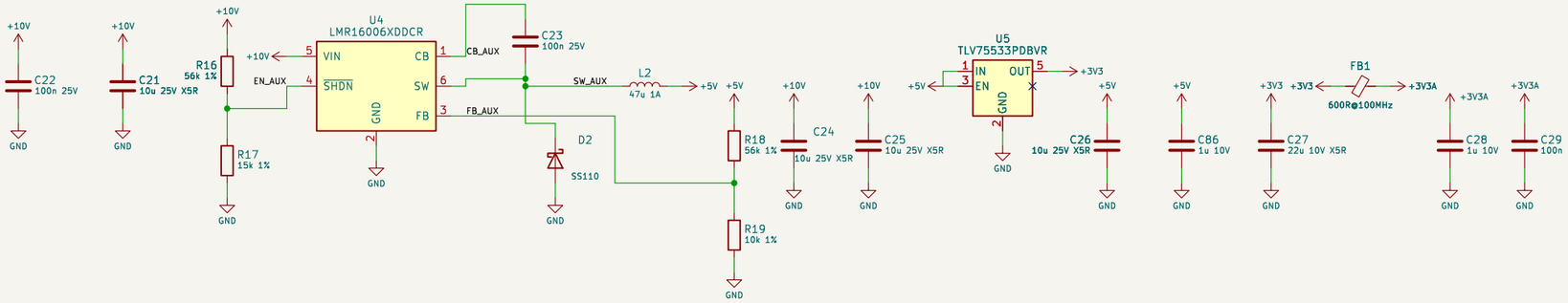


Sheet 5: 4SBB regulator — power stage and gate drivers
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

Sheet: /4SBB regulator/
File: sbb.kicad_sch

Title: 4SBB regulator — power stage and gate drivers

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/10



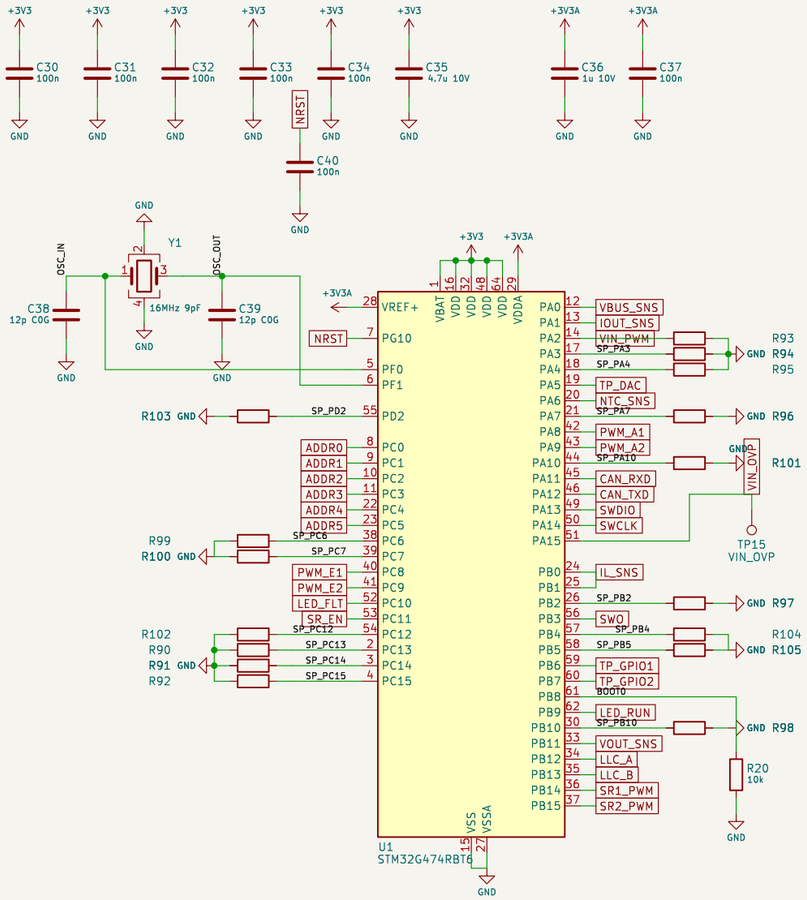


Sheet 6: Secondary aux supplies
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

Sheet: /Secondary aux supplies/
File: sec_aux.kicad_sch

Title: Secondary aux supplies

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/10



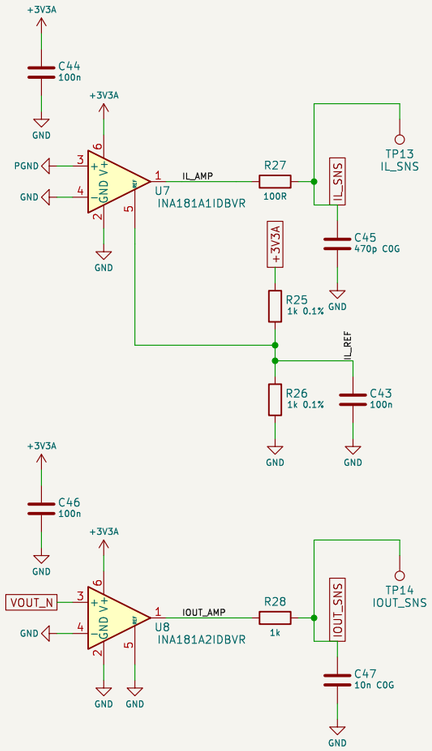
Unused GPIOs (PC13–15, PA3–4, PA7, PA10, PB2, PB4, PB5, PB10, PC6, PC7, PC12, PD2):
10k pull-down to GND each (R90–R105). PA4 is kept free for DAC1_OUT1 (Test_pin_DAC).
PA2 = VIN_PWM (TIM15_CH1 PWM input, VIN monitor from the primary); R93 is its pull-down.

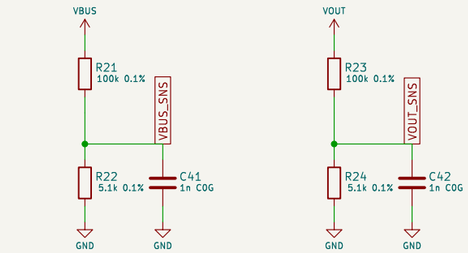
Sheet 7: STM32G474RBT6 MCU
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

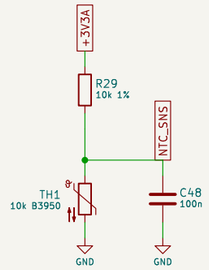
Sheet: /STM32G474RBT6 MCU/
File: mcu.kicad_sch

Title: STM32G474RBT6 MCU

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/10





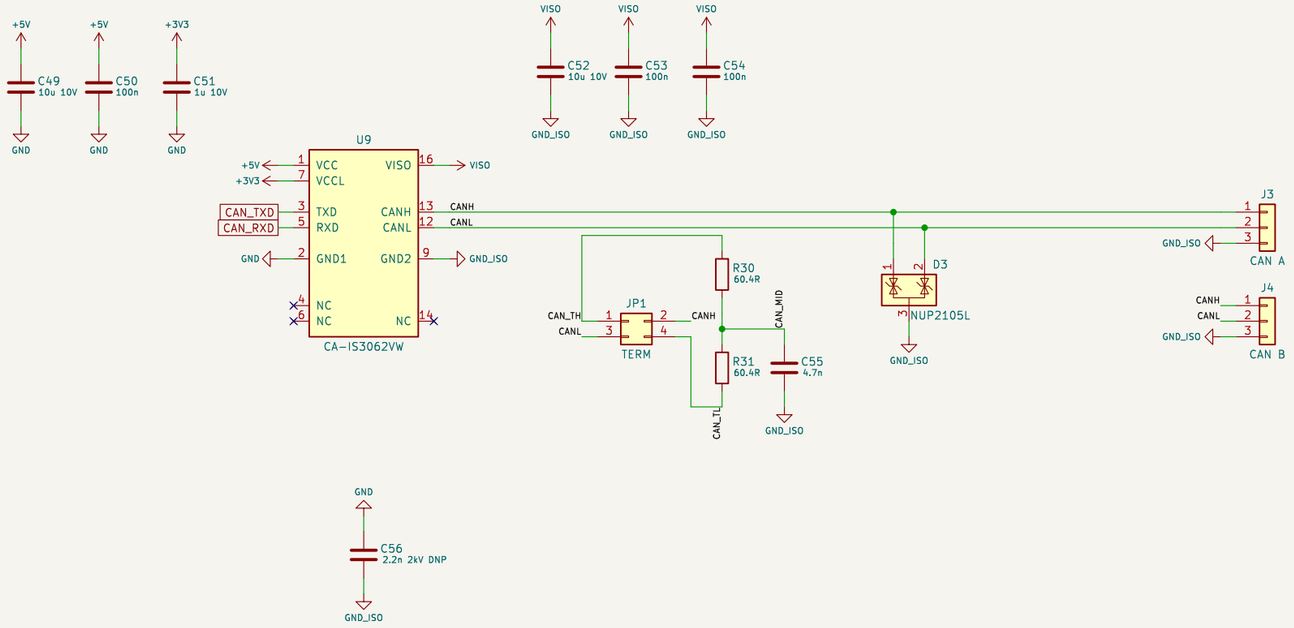


Sheet 8: Sensing – VBUS, VOUT, inductor and output current, NTC
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

Sheet: /Sensing/
File: sense.kicad_sch

Title: Sensing – VBUS, VOUT, inductor and output current, NTC

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/10

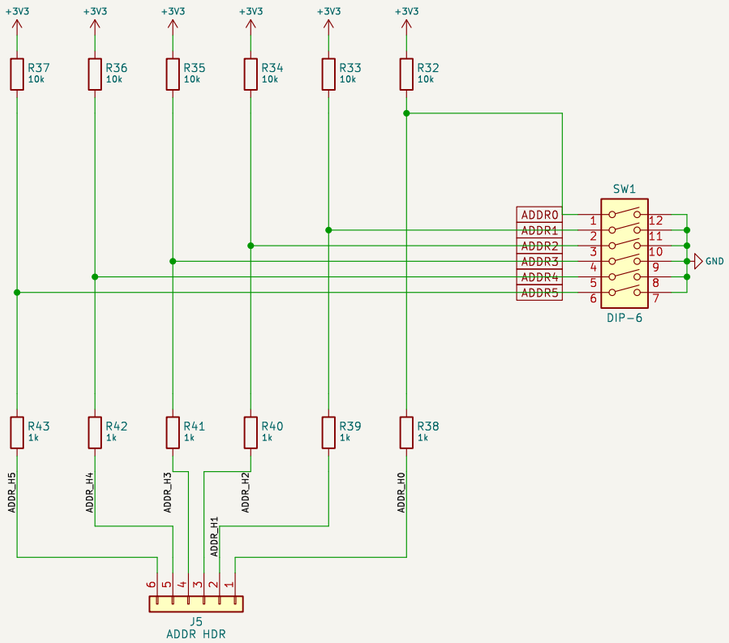


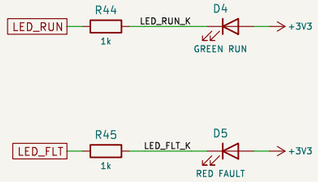
Sheet 9: Isolated CAN
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

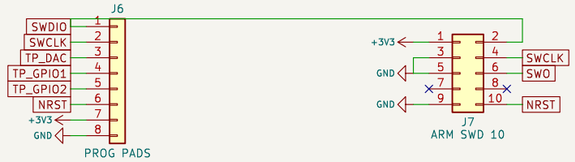
Sheet: /Isolated CAN/
File: can.kicad_sch

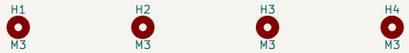
Title: Isolated CAN

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/10









Sheet 10: Address switch, programming pads, ARM SWD, LEDs
DCDC LLC + 4SBB, 12–48 V in, 1.5–48 V / 5 A out, STM32G474

Sheet: /Address switch, programming pads, ARM SW/
File: io.kicad_sch

Title: Address switch, programming pads, ARM SWD, LEDs

Size: A3	Date: 2026-10-19	**Rev: E**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/10